

The Computational Boundary of Scalar Temporal Attention: A Mechanistic Analysis of the Temporal Attention Neuron

Li Zexu

School of Physics and Astronomy, University of Leeds

Correspondence: [email placeholder — see archive metadata]

Abstract

Background. When attention-like temporal weighting is compressed into a single dynamical unit, its true computational capabilities—as opposed to its architectural resemblance to attention—remain unclear. **Methods.** We analyse the Temporal Attention Neuron (TAN), a scalar-input spiking neuron with a windowed, surprise-gated, Boltzmann-style temporal weighting kernel, using four frozen experimental programmes: a natural state-collision test of the Markovianity of its membrane potential (Phase 1); event-locked response geometry measured by effective rank, log-trace and spectra of pooled response covariances (Phase 2); a retrieval-cue temporal distractor task (Probe 1); and an identifiability audit of query-conditioned binding (Probe 2), with pre-registered gates, counterfactuals and calibration audits throughout. **Results.** TAN is history-dependent and its scalar membrane potential is not a sufficient state; surprise-gated attention reorganizes the event-locked local response geometry relative to uniform temporal integration, although the shared-coordinate geometry shows no TAN-specific advantage over LIF. Probe 1 provides constrained, seed-attenuated evidence of target-relevant retrieval relative to the no-attention control. Probe 2 fails every routing gate: with the frozen kernel the query enters attention as a common multiplicative surprise factor, so changing the query never changes which historical item is selected (counterfactual attention-argmax flip rate 0.000). **Conclusions.** Scalar temporal attention provides dynamic, surprise-conditioned saliency weighting, but not genuine query-conditioned content-addressable routing; temporal memory, attention-like weighting, routing and binding are distinct computational properties.

Keywords: temporal attention; spiking neuron; non-Markovian dynamics; state sufficiency; effective rank; mechanistic identifiability; routing; binding

1 Introduction

Attention has become a general computational primitive. In machine learning, content-based weighting over key–value pairs underlies sequence transduction (Vaswani et al., 2017; Bahdanau et al., 2015; Luong et al., 2015), pretrained language models (Devlin et al., 2019; Brown et al., 2020) and is studied mechanistically as a form of information routing (Elhage et al., 2021; Olsson et al., 2022). In neuroscience, attention-like gain and selection have been modelled at the level of normalization (Reynolds and Heeger, 2009), saliency (Koch and Ullman, 1985; Itti et al., 1998) and dynamic routing of information (Olshausen et al., 1993), while binding has classically been associated with temporal correlation and synchrony (von der Malsburg, 1995; von der Malsburg and Schneider, 1986; Singer and Gray, 1995; Treisman and Gelade, 1980).

Most of these mechanisms are studied at the population or network level. This raises a more primitive question that is rarely asked: *what happens if attention-like temporal weighting is compressed into a single dynamical unit?* A single neuron is a scalar dynamical system with a leaky membrane potential; classical integrate-and-fire dynamics (Lapicque, 1907; Abbott, 1999; Hodgkin and Huxley, 1952; Burkitt, 2006; Gerstner et al., 2014) have no mechanism for query-dependent selection of stored content. Whether a minimal single-neuron architecture can nevertheless exhibit history-dependent computation, and where its mechanistic boundary lies, is an open question about the nature of attention as a computational primitive rather than about a specific benchmark.

The Temporal Attention Neuron (TAN) provides a minimal laboratory for this question. TAN is a discrete-time single neuron that combines (i) a temporal receptive field over the recent input stream, (ii) a surprise-gated, Boltzmann-style weighting of that window – an attention-like saliency kernel – and (iii) leaky spiking membrane dynamics. The presence of attention-like weighting inside a dynamical unit does not, however, guarantee that the unit can *route* information: weighting strength is not content selection.

The purpose of this paper is therefore not to propose a better neuron, but to perform a mechanistic decomposition of one minimal construction. We ask, in sequence, four nested questions. (i) *State*: does temporal history become part of the effective state, so that the scalar membrane potential alone is insufficient? (ii) *Geometry*: does attention-like weighting reorganize the local, event-locked response geometry relative to uniform temporal integration? (iii) *Computation*: does this history contribute to target-relevant retrieval in a controlled temporal task? (iv) *Identifiability*: does the scalar attention kernel actually implement query-conditioned routing, i.e. a query determining which stored event is read out?

These capabilities form a hierarchy that must be verified layer by layer:

$$\text{memory} \rightarrow \text{weighting} \rightarrow \text{routing} \rightarrow \text{binding}.$$

Each arrow is a separate computational claim. The experiments reported here establish the first two layers for TAN, provide limited evidence on the third, and show that the fourth is not implemented by the current scalar kernel. The central thesis is:

A minimal temporal-attention neuron can acquire history-dependent, non-Markovian dynamics and can reorganize event-locked response geometry through surprise-gated temporal weighting; however, its scalar attention kernel does not provide genuine query-conditioned content-addressable routing. Temporal memory, attention-like saliency weighting, routing and semantic binding are therefore distinct computational capabilities, not interchangeable consequences of increasing representational complexity.

The remainder of the paper is organized as follows. Section 2 positions TAN with respect to state sufficiency, input-dependent weighting, routing and mechanistic identifiability. Section 3 gives the frozen model definition. Section 4 describes the experimental program (state-sufficiency collisions, event-locked geometry, a temporal distractor task, and an identifiability audit). Section 5 reports results; Section 6 interprets them; Section 7 lists limitations; Section 8 concludes with the conceptual separation that this paper establishes.

2 Related Concepts and Theoretical Background

2.1 Temporal memory and state sufficiency

A dynamical system is Markovian in a state variable if that variable summarizes all information from the past that is relevant to the future. For leaky integrate-and-fire (LIF) neurons (Lapicque,

1907; Abbott, 1999; Burkitt, 2006), driven by an external scalar input, the membrane potential together with the input history constitutes the state; with a fixed input schedule the membrane potential alone is a sufficient statistic in the sense that identical potentials with identical future inputs produce identical futures. Working-memory models at the population level instead maintain content through attractor or delay activity (Goldman-Rakic, 1995; Durstewitz et al., 2000; Compte et al., 2000; Hopfield, 1982), while recurrent sequence models exhibit temporal state through hidden dynamics (Elman, 1990; Maass et al., 2002; Jaeger, 2001). For a single neuron, whether history enters the transition law at all, and whether the scalar potential remains a sufficient projection, is an empirical question that a state-collision experiment can answer directly: two histories that lead to (nearly) identical membrane potentials are injected with an identical probe; if the next-state potentials diverge, the scalar potential is not a sufficient state and the projection onto it is not a consistent (lumpable) reduction (Kemeny and Snell, 1960).

2.2 Attention as input-dependent weighting

Attention in sequence models computes a normalized, input-dependent weighting of values by query-key compatibility (Vaswani et al., 2017; Bahdanau et al., 2015); the weights form a Gibbs distribution, connecting such weighting to statistical mechanics (Jaynes, 1957; Gibbs, 1902). In sensory neuroscience, saliency weighting describes bottom-up prominence of inputs (Koch and Ullman, 1985; Itti et al., 1998), and normalization models describe input-dependent gain (Reynolds and Heeger, 2009). Neuromodulatory gain control offers a biological analogue of input-conditioned modulation (Aston-Jones and Cohen, 2005; Yu and Dayan, 2005). Predictive-coding accounts treat deviation-from-expectation signals as the driver of cortical gain (Rao and Ballard, 1999; Srinivasan et al., 1982; Friston, 2005; Keller and Mrsic-Flogel, 2018; Bastos et al., 2012), and single-cell asymmetric response properties are natural substrates for such deviations (Laughlin, 1981). TAN instantiates input-dependent weighting as a surprise-scaled, Boltzmann-style reweighting of its own temporal window; the key theoretical property we later exploit is that the surprise variable is a scalar that multiplies every window entry uniformly.

2.3 Routing and content-addressable selection

Routing is the ability to direct information from a selected source to a consumer based on control signals. Classical proposals in vision route attended locations into an invariant representation (Olshausen et al., 1993), mixture-of-experts models route inputs to experts through a gating network (Jacobs et al., 1991; Shazeer et al., 2017; Fedus et al., 2022), and mechanistic analyses of transformers interpret attention heads as content-addressable key-value lookups that implement induction and copying behaviour (Elhage et al., 2021; Olsson et al., 2022). Binding – linking a selected content to a role or query – has been associated with feature-integration (Treisman and Gelade, 1980), correlation/synchrony (von der Malsburg, 1995; von der Malsburg and Schneider, 1986; Singer and Gray, 1995; Fries, 2005), and, in cellular terms, with dendritic and circuit mechanisms (London and Häusser, 2005; Larkum, 2013; Spruston, 2008). An important interpretive caution applies to all of these literatures: observed weight values are not by themselves demonstrations of routing or explanation (Jain and Wallace, 2019; Serrano and Smith, 2019; Wiegrefe and Pinter, 2019); a weighting mechanism must be shown to change *which content* is selected as a function of a control signal, not merely how strongly a fixed content is weighted.

2.4 Mechanistic identifiability

When a task can be solved through unintended shortcuts – fixed positional memory, amplitude cues, or query–label correlations – a measured advantage of one architecture over another cannot be attributed to the mechanism under study. Mechanistic identifiability requires that (i) simpler storage/integration mechanisms cannot solve the task through linear readouts, (ii) the mechanism of interest is observable (e.g. the attention weights themselves), and (iii) counterfactual manipulations (permuting positions, values, queries) change the mechanism in the direction predicted by the scientific claim. This paper applies this audit discipline to every layer of the hierarchy above; negative audit outcomes are treated as mechanistic results, not as experimental failures.

3 The Temporal Attention Neuron

3.1 Definition

TAN is a discrete-time single-neuron model. At time t the neuron maintains a temporal receptive field over the last W inputs,

$$\mathcal{X}_t = [x_{t-W+1}, x_{t-W+2}, \dots, x_t], \quad (1)$$

with $x_t \in \mathbb{R}$ a scalar input. The local mean

$$\mu_t = \frac{1}{W} \sum_{i=1}^W x_{t-W+i} \quad (2)$$

defines the rectified surprise (prediction-error-like deviation),

$$S_t = [x_t - \mu_t - \varepsilon]_+, \quad (3)$$

where $[\cdot]_+$ denotes rectification and $\varepsilon \geq 0$ is a noise tolerance. A query, key and value are obtained by linear maps of the surprise and of the window,

$$Q_t = W_q S_t, \quad K_{t,i} = W_k x_{t-W+i}, \quad V_{t,i} = W_v x_{t-W+i}, \quad (4)$$

and Boltzmann attention logits and weights are

$$E_{t,i} = Q_t K_{t,i}, \quad \alpha_{t,i} = \frac{\exp(\beta E_{t,i})}{\sum_{j=1}^W \exp(\beta E_{t,j}) + 10^{-9}}. \quad (5)$$

The attention-weighted context is

$$C_t = \sum_{i=1}^W \alpha_{t,i} V_{t,i}, \quad (6)$$

and the gated synaptic current is

$$A_t = \tanh(S_t) C_t. \quad (7)$$

The membrane potential evolves as

$$h_t = \lambda h_{t-1} + A_t, \quad (8)$$

with the spiking readout

$$y_t = \Theta(h_t - \theta), \quad h_t \leftarrow 0 \text{ if } y_t = 1. \quad (9)$$

The frozen parameters used throughout this paper are $W = 5$, $\lambda = 0.5$, $\theta = 0.5$, $W_q = 2.0$, $W_k = W_v = 1.0$, $\beta = 1.0$, $\varepsilon = 0.0$, with the 10^{-9} softmax offset. The receptive field is zero-padded at trial onset and $h_0 = 0$. The architecture and information flow are summarized in Figure 1. All model parameters, conventions and implementation parity checks are recorded in the frozen research archive accompanying this paper (see Data and Code Availability), and the reference implementation is `code/model/tan.py` in that archive.

3.2 Effective state and projection

Because the input stream is scalar and the receptive field is a finite window, the natural state of TAN is the pair

$$\mathbf{s}_t = (h_t, \bar{X}_t), \quad \bar{X}_t = (x_{t-W+2}, \dots, x_t), \quad (10)$$

together with the readout convention, and the scalar membrane potential is the projection $\pi(h, \bar{X}) = h$. Whether π is a sufficient (lumpable) reduction is an empirical question that Phase 1 (Section 4.2) addresses with a natural-collision protocol: if two histories with π -identical states ($h_{T-1}^{(A)} \approx h_{T-1}^{(B)}$) can be driven apart by one identical probe, the transition law depends on more than h and the projection is not lumpable (Kemeny and Snell, 1960).

3.3 Surprise-gated temporal weighting is scalar

A structural property of the frozen kernel is central to this paper. At a fixed time step the attention logits factorize as

$$E_{t,i} = \beta W_q W_k S_t x_{t-W+i} = \gamma_t x_{t-W+i}, \quad \gamma_t = \beta W_q W_k S_t \in \mathbb{R}, \quad (11)$$

that is, a common scalar γ_t times the window values. Whenever $S_t > 0$ (hence $\gamma_t > 0$),

$$\arg \max_i E_{t,i} = \arg \max_i x_{t-W+i}, \quad (12)$$

so the attention ranking of history entries is determined by their amplitudes, and the current surprise only sets a common temperature. A query that enters the model *only through* S_t can therefore change attention sharpness but not the relative ordering of historical entries. This factorization is the theoretical precheck behind Probe 2 (Section 4.5): with the frozen kernel, query-conditioned reordering of history is structurally excluded as long as the query acts only through the scalar surprise.

3.4 Comparison models

Four models share the same spiking readout ($\theta = 0.5$, $\lambda = 0.5$, reset on spike) and the same scalar input:

- **B1 (LIF):** $h_t = \lambda h_{t-1} + x_t$; no window use.
- **B2 (LIF + fixed delay line):** $h_t = \lambda h_{t-1} + \mathbf{w}^\top \mathcal{X}_t$ with fixed recency weights $w_i = i / \sum_j j$; passive, position-invariant linear memory.
- **B3 (TAN without attention):** uniform weights $\alpha_{t,i} = 1/W$, so $C_t = \mu_t$; surprise gating retained.
- **B4 (Full TAN):** the model of Section 3.1.

A no-dynamics control (**C0**) consists of input-derived features only. All implementations are equation-identical to the reference model code; vectorised experiment implementations are parity-checked against the scalar reference implementation (worst deviation 1.5×10^{-10} in h ; spike agreement exact).

4 Experimental Program

4.1 Overview

The program follows the hierarchy of Section 1: state sufficiency (Phase 1), local fluctuation geometry (Phase 2), target-relevant retrieval (Probe 1), and identifiability of query-conditioned routing (Probe 2). All experiments, seeds, parameters, audit rules and results are frozen and recorded in the accompanying research archive; this paper reports only archived results. Statistical procedures are summarized in Section 4.6.

4.2 Phase 1: state sufficiency

Question. Is the scalar membrane potential h_t a sufficient Markov state of TAN?

Natural-collision protocol. $N = 160,000$ random histories of length $T - 1 = 9$ were drawn from four stimulus families (Gaussian white noise, random steps, ramps, sine waves). For each model separately, all unordered pairs (A, B) whose subthreshold potentials at $t = T - 1$ collide – $0.05 < h_{T-1} < \theta$ on both sides and $|h_{T-1}^{(A)} - h_{T-1}^{(B)}| < \delta = 10^{-4}$ – form the *natural-collision ensemble*. Restricting to the subthreshold band excludes trivial reset coincidences ($h = 0$). At $t = T = 10$ an identical probe $x^* \sim \mathcal{U}(0.5, 3.0)$ is injected into both histories and the one-step divergence is measured on the continuous branch (pre-reset potential $u_T = \lambda h_{T-1} + A_T$), i.e.

$$\Delta \tilde{h}_T = |u_T^{(A)} - u_T^{(B)}|, \quad K = \Delta \tilde{h}_T / |h_{T-1}^{(A)} - h_{T-1}^{(B)}|. \quad (13)$$

Post-reset divergence, spike-outcome classes and reset re-synchronisation (a common double spike at T , then identical continuation inputs) are tabulated separately. A Markovian claim predicts $K \leq 1$ -like contraction; LIF predicts $K = \lambda$ exactly on every collision pair.

4.3 Phase 2: local fluctuation geometry

Question. Does surprise-gated weighting reorganize the local, event-locked response geometry relative to uniform temporal integration?

Protocol. Sparse episodic stimuli (Poisson-like onset gaps with burst structure and amplitude diversity; $T = 5000$, burn-in 500, three fixed seeds) drive B1–B4 and an input-only control C0. For each model a response trajectory is embedded in frozen coordinate frames: $z_C = [h, \mu, S, A]$ for all models (B1’s A is its input, a null convention), and $z_F = [h, \mu, S, A, \alpha_1, \dots, \alpha_5, C]$ for B3/B4 (matched frame). Coordinates are globally Z -scored per seed (post-burn-in) and per-seed centred before cross-seed pooling; no whitening is applied. Events are aligned at episode onset and analysed at fixed lag $\tau \in [-6, 30]$; the primary response is the baseline-subtracted state $\Delta z_{e,\tau}$ relative to a five-step pre-event window, pooled across events (never per-window PR averaging). At each lag we compute the effective rank of the pooled response covariance,

$$d_G(\tau) = \frac{\text{Tr}(C_z)^2}{\text{Tr}(C_z^2)}, \quad (14)$$

its log-trace $\log \text{Tr}(C_z)$ and eigenvalue spectrum, and the velocity geometry

$$d_D(\tau) = \frac{\text{Tr}(C_v)^2}{\text{Tr}(C_v^2)}, \quad v_{e,\tau} = z_{e,\tau+1} - z_{e,\tau}, \quad (15)$$

where C_v is the pooled covariance of same-event adjacent-phase increments. d_{PR} is reported and interpreted strictly as an *effective rank of the local fluctuation covariance*; it is never called an intrinsic or manifold dimension (Gao and Ganguli, 2015; Mazzucato et al., 2016; Facco et al., 2017; Chung and Abbott, 2021). Zero-variance windows return NaN (never a fabricated value). Amplitude controls are computed for every lag: raw pooled, amplitude-stratified (median split on the log-amplitude, pooled-within covariance) and per-coordinate residualisation on log-amplitude. Quiet-reference samples lie ≥ 12 steps from any pulse. Recovery times ($\tau_{1/2}$ on d_D , $\tau_{e\text{-fold}}$ on $\log \text{Tr}$) are estimated on isolated events with event bootstrap confidence intervals.

4.4 Probe 1: temporal distractor task

Question. Does stored history contribute to target-relevant retrieval, as opposed to mere storage?

Task (frozen v3). Each trial has five event steps: a target at fixed step 5 whose amplitude class c_T is low ($\mathcal{U}(0.55, 1.05)$) or high ($\mathcal{U}(0.95, 1.45)$) with overlapping supports; either three distractors at steps 6–8 (drawn from the same class mixture, so a distractor may exceed the target) or silence; and a query at step 9 that acts as a *retrieval cue* but does not enter the label. The label is three-way: $c_T \in \{\text{lo}, \text{hi}\}$ or *absent* (catch, one third of trials). The query re-opens the surprise gate so the window (which still contains the target at step 9) can be re-read into the response. A linear three-class softmax readout is trained on response-trajectory features $F = [u_9, \dots, u_{13}, y_9, \dots, y_{13}]$; per-step observables before the query are never visible (no amplitude leakage at arrival). Readout protocol, training (clean + distractor mixed, $n_{\text{train}} = 4000$), testing ($n_{\text{test}} = 2000$) and the three seeds are identical across B1–B4. Primary endpoint: $\Delta_{\text{dist}} = \text{acc}(\text{clean}) - \text{acc}(\text{distractor})$; main contrast B4 vs. B3.

4.5 Probe 2: query-conditioned binding identifiability audit

Question. Does the current query determine which historical event is selected ($Q \rightarrow K \rightarrow V$)?

Audit design. Two events $E_A = (K_A, V_A)$, $E_B = (K_B, V_B)$ with key amplitudes $K_A = 1.0$, $K_B = 2.0$ and values $V \in \{-1, +1\}$ (fair) are encoded into the scalar stream as $x(E) = K_{\text{amp}} + V\Delta$, $\Delta = 0.25$; the two events occupy two uniformly random history slots (of four), the remaining slots carry noise $\mathcal{N}(0, 0.05^2)$, and the query is $x_t = 3.0$ ($q = A$) or 4.0 ($q = B$). The answer is $y = V_q$. Identifiability gates: (A) a linear readout on $z = [x_{t-4}, \dots, x_t]$ (B2 shortcut test) must not exceed 0.58 balanced accuracy; (B) the same readout on B3’s uniform-integration response features must not exceed 0.58; (C) for B4, attention must hit the target history position in $\geq 50\%$ of trials (chance 25%), show target/distractor contrast $R = \mathbb{E}[\alpha_{\text{target}}] / \mathbb{E}[\alpha_{\text{distractor}}] \geq 2$, and produce query-dependent attention (Jensen–Shannon divergence between $\alpha | q=A$ and $\alpha | q=B$ distributions with permutation $p < 0.01$). Counterfactual audits permute positions, values and queries, and hold history fixed while changing the query. All gates, thresholds and the final status are frozen: **TERMINATED** / **AUDIT_FAILED** is the archived outcome of this audit, i.e. an identifiability boundary result, not an incomplete experiment.

4.6 Statistical and reproducibility procedures

All experiments use fixed seeds and frozen parameters recorded in the archive (docs/REPRODUCIBILITY_LEDGER.md). Confidence intervals are event/trial bootstraps (2000

resamples) unless stated. Calibration audits (generator sanity, hidden-label checks, shuffled-label readouts, class-coverage, permutation nulls for shuffle statistics) are executed before formal runs; violations stop the run. Probe-1’s three-seed data were re-analysed after an audit-statistics amendment (empirical permutation null for the shuffle audit; per-cell coverage statuses instead of run-blocking flags) on the frozen data alone, with the deterministic reproduction verified against the frozen per-seed numbers (maximum deviation 4.7×10^{-4} , rounding only). No result in this paper was obtained by post-hoc tuning.

5 Results

All numbers in this section are frozen archive results (Section 4); each claim is traceable to a recorded experiment.

5.1 History dependence and non-Markovianity

Natural collisions were abundant for every window-carrying model (found pairs: B1 92,600; B2 19,234; B3 1,633,844; B4 1,427,817; 1,500 analysed per model; Figure 2, Table 2). On the continuous branch, the mean one-step amplification K was 0.5000000 for B1 (exact; maximum deviation 1.8×10^{-8} , machine precision), 4.47×10^3 (95% CI $[3.26, 5.92] \times 10^3$) for B2, 1.235×10^4 ($[7.78, 1.83] \times 10^4$) for B3 and 2.047×10^4 ($[1.12, 3.57] \times 10^4$) for B4, with 73.7% of B4 collision pairs exceeding $K > 10^3$. Post-reset Markov violations ($|\Delta h_{10}| > \delta$) occurred in 0%, 0.6%, 33.1% and 32.2% of pairs for B1–B4 respectively. When the probe was not surprising relative to the local window (gates closed, $S_{10} = 0$), the current vanished and $K = 0.5$ exactly for both B3 and B4: surprise-gating makes the non-Markovianity *conditional* on prediction error. Attention divergence predicted the current divergence (Pearson $r(JSD, \Delta A) = 0.87\text{--}0.89$, $p < 10^{-300}$; ΔA vs. $\tilde{\Delta h}$ $r = 1.000$), and the amplification ordering was $B4 > B3 > B2$ (median K ratios 2.15 and 3.29, Mann–Whitney $p = 1.3 \times 10^{-32}$ and $p = 1.3 \times 10^{-63}$). After a common double spike at T (843 fully resynchronised pairs), B1 stayed synchronised forever (machine-exact), B2 re-diverged on the continuous branch and re-converged after the window flush, while B3/B4 retained inter-pair gaps beyond the flush in $\sim 25\%$ of pairs: a spike reset erases h but not the windowed memory. These results establish that the scalar membrane potential is not a sufficient state whenever the update reads the windowed past, for *all* window-carrying models; history dependence is not an attention-specific effect.

5.2 Local fluctuation geometry

With the shared four-dimensional frame z_C and a matched full frame z_F (Section 4.3; Figure 3, Figure 4, Table 3), event-locked velocity effective rank d_D (mean over $\tau \in \{0, 1, 2\}$, 374 pooled events) was: B1 1.674 (Δ vs. quiet 0.674), B3 1.458 (z_C) and 1.484 (z_F), B4 1.589 (z_C) and 2.089 (z_F ; $\Delta = 1.089$), and the input-only control C0 1.055. Three findings follow. First, event-locked expansion exists for every h-bearing model and is not specific to TAN: in the shared frame B4 does not exceed B1 ($1.589 < 1.674$), and this negative contrast is robust across amplitude controls (raw/stratified/residualised 1.589/1.701/1.712 vs. B1 1.674/1.732/1.747). Second, within the matched full frame that contains the attention coordinates, B4 exceeds B3 in effective rank (event d_D 2.089 vs. 1.484; event CIs $[1.87, 2.30]$ vs. $[1.42, 1.53]$, disjoint) and in covariance scale (event $\log \text{Tr}$ 3.58 vs. 2.48 nats; $\Delta = +1.09$ nats); the rank contrast survives and slightly strengthens under amplitude controls (+0.61 raw, +0.74 stratified, +0.78 residualised). Third, the velocity-geometry response is temporally structured: its peak occurs at $\tau \approx 4\text{--}6$ (content-rich windows after burst onsets, d_D up to 2.89), not at the first pulse ($d_D(\tau=0) \approx 1.66$), and recovery is faster for B4

($\tau_{e\text{-fold}} = 5.4$, CI [4.6, 6.3], vs. B3 8.5, [6.1, 10.6]). These are statements about the effective rank of local response covariance; they are not statements about intrinsic dimension.

5.3 Target-relevant temporal retrieval (Probe 1)

Pooled balanced three-class accuracy over the frozen three seeds (Figure 5, Table 4) was, for clean / distractor conditions: B1 0.331/0.340 (chance), B2 0.867/0.843 (strong, stable positional-memory baseline), B3 0.375/0.337 (with lo-class-blind cells; one cell invalid for interpretation), B4 0.328/0.389. The B4 vs. B3 contrast in the distractor condition was +0.0570 (95% CI [+0.0387, +0.0758], $n = 6000$ pooled trials), with per-seed differences +0.056, +0.043, +0.056. B4’s absolute elevation over chance in the distractor condition attenuated across seeds (0.427, 0.398, 0.342; seed 20260906 is $\approx$ chance). Taken together, and with B3’s readout limitation and the task’s structural bias toward fixed-position memory (B2 dominates), the evidence supports the constrained statement: Full TAN shows evidence of improved target-relevant retrieval under content-rich temporal context relative to the no-attention control, with a partially replicated, seed-attenuated effect. This is *limited support*, not a demonstration of flexible temporal retrieval or distractor robustness.

5.4 Identifiability audit of query-conditioned binding (Probe 2)

The identifiability audit did not support genuine query-conditioned routing in the current scalar TAN kernel (Figure 6, Table 5). The B2 linear shortcut test reached balanced accuracy 0.779 ([0.724, 0.836]) and the B3 shortcut test 0.709 ([0.644, 0.775]) against a red line of 0.58, so the task as encoded was solvable through storage/integration shortcuts before any routing mechanism was evaluated. For B4, the attention hit rate (argmax of the history-tap attention weights landing on the target position) was 0.511 (bootstrap CI lower bound 0.481), split as 0.000 for $q = A$ and 1.000 for $q = B$; the target/distractor contrast was $R = 0.034$ (required ≥ 2); and the query-conditioned JSD of attention distributions was 0.0002 with permutation $p = 0.92$ (no detectable query dependence). The decisive counterfactual: with history held fixed and only the query changed ($q = A \rightarrow B$), the attention argmax flip rate was 0.000. Changing the query changed attention sharpness, never the winner among history taps.

5.5 Mechanistic diagnosis

Three layers explain the audit outcome (Figure 7). *Layer 1, encoding shortcut*: the composite encoding used asymmetric key amplitudes, so B-coded events were systematically larger; storage/integration readouts could exploit static amplitude, and B4’s amplitude-based attention was attracted to B events ($q = B$ hit 1.000, $q = A$ hit 0.000). *Layer 2, scalar-kernel limitation*: independently of the encoding, with $E_{t,i} = \gamma_t x_i$ the ratio $E_{t,i}/E_{t,j} = x_i/x_j$ is query-independent for fixed history; query identity enters only through the scalar γ_t , so query changes weighting strength, not selected content. *Layer 3, self-attraction*: query amplitudes (3.0/4.0) exceeded all history codes, so the softmax concentrated on the current tap in 100% of trials – an experimental confound that does not explain away the structural limitation of the scalar kernel.

6 Discussion

6.1 What scalar temporal attention does compute

Taken together, the frozen results characterize TAN as

prediction-error gating + history dependence + scalar temporal saliency weighting.

The state-collision experiments establish history dependence and show it to be conditional on surprise: in gate-closed episodes the dynamics collapse to exact LIF contraction, and in gate-open episodes the windowed context is re-coupled into the membrane through an attention-weighted current. The geometry experiments show that this re-coupling has an observable, event-locked signature in the effective rank and covariance scale of the response representation, and that adaptive attention adds structure beyond uniform temporal integration in the frame that contains the attention coordinates. The retrieval experiment provides constrained evidence that such structure can support target-relevant readout in content-rich windows, relative to the no-attention control, in a task whose fixed-position layout structurally favours a passive delay line (B2).

6.2 What scalar temporal attention does not compute

The same body of evidence shows what the minimal kernel does not provide. In the shared observable coordinates, TAN does not possess a higher effective rank of event responses than LIF; effective-rank expansion is generic to history-carrying units. On a fixed-position retrieval task, a passive delay line outperforms attention variants. And in the identifiability audit, the scalar kernel failed every routing gate: attention focus did not follow the query, the target/distractor contrast was inverted, and holding history fixed while switching the query never changed which history tap was attended. The architecture therefore does not implement content-addressable Q - K routing, and semantic binding is not established by any result in this paper.

6.3 Why attention weighting is not routing

The mechanistic reason is the factorization of Section 3.3: with a scalar surprise S_t , attention logits are a common scalar times the window values, so their *relative ordering* is a property of the input amplitudes alone. A query that enters only through S_t can modulate how sharply the distribution concentrates – a gain-like, saliency-like operation – but cannot determine *which* historical item is selected. Weighting strength and content selection are therefore different operations, consistent with the interpretive literature on attention weights in general (Jain and Wallace, 2019; Serrano and Smith, 2019; Wiegrefe and Pinter, 2019): observed weights can be read as explanations or routing decisions only if the mechanism is shown to change selected content under a control signal.

6.4 Representation complexity is not computational selectivity

A recurrent theme across phases is that richer state structure does not by itself confer selective computation. Phase 2 showed representational reorganization in attention-bearing coordinates; Probe 1 showed that a fixed memory baseline can outperform attention on a storage-aligned task; Probe 2 showed that a scalar weighting mechanism with nontrivial history dependence cannot select content by query. These results caution against inferring computational capability from state or representation complexity alone (Chung and Abbott, 2021; Gao and Ganguli, 2015), and against treating dimensionality-like statistics as intrinsic capacity measures (Facco et al., 2017). We make no complexity theorem here; the empirical point is that the layers of the hierarchy in Section 1 must each be verified directly.

6.5 Relation to dynamical systems

From a dynamical-systems perspective, TAN is a non-autonomous, input-driven map whose contraction is conditional on a prediction-error gate: leaky decay guarantees boundedness (Lohmiller and Slotine, 1998; Strogatz, 2014), while windowed updates make the effective state higher-dimensional

and the one-step divergence of near-colliding states reflect the local expansion of the driven map. Reset re-bifurcation shows that the spiking readout does not erase the windowed memory, connecting the single-neuron setting to delay- and attractor-based memory accounts (Compte et al., 2000; Durstewitz et al., 2000; Elman, 1990). Standard nonlinear time-series tools (embedding, effective dimension) (Takens, 1981; Grassberger and Procaccia, 1983; Kennel et al., 1992; Facco et al., 2017; Bandt and Pompe, 2002) motivated our observable choices, but we report only estimator-level statistics (effective rank, log-trace, spectra) with the caveats stated in Section 7.

6.6 Implications for multi-neuron architectures

The mechanistic boundary of the scalar kernel identifies what an organizing principle would need to add: a mechanism in which a query can change the *relative* compatibility of candidate memories. Candidate organizational directions – distinct from the present results – include multiple dendritic compartments (London and Häusser, 2005; Larkum, 2013; Spruston, 2008), opponent/excitatory–inhibitory circuits, lateral inhibition and competitive microcircuits, and phase- or synchrony-based binding (von der Malsburg, 1995; Singer and Gray, 1995; Fries, 2005). All such directions are future hypotheses; none is tested or claimed here.

7 Limitations

The results in this paper are subject to explicit limitations, recorded as part of the frozen archive rather than resolved post hoc.

1. **Scalar input.** All experiments use single-channel, scalar stimulus streams; no multimodal or vector-valued input was studied.
2. **Small window.** $W = 5$; conclusions about history dependence concern a very short receptive field.
3. **Simplified attention kernel.** A single scalar query with a linear key map; the kernel is a deliberate minimal construction, and its boundaries (Section 5.5) are therefore properties of this construction.
4. **Synthetic tasks.** Stimuli are synthetic (noise, steps, ramps, pulses, episodes); no naturalistic or behavioural data were used.
5. **Probe-1 readout limitations.** The linear readout is a frozen protocol constraint; readout class-collapse occurred in some cells and is handled by per-cell audit statuses, not by removing data.
6. **Probe-1 B3 class coverage.** B3 (no attention) exhibited lo-class-blindness in its readout in one seed (clean condition, bal3 0.470 marked invalid for interpretation) and near-chance, partially degenerate behaviour elsewhere; B3 is therefore not a clean computational baseline for Probe-1, and contrasts involving it must be read with this caveat.
7. **Probe-2 encoding leakage.** The composite encoding carried conditional single-tap information at B-coded amplitudes ($|\mathbb{E}[V_q | x_i]|$ up to ~ 0.5 – 0.6 at code values); this is recorded as an experimental confound of the audit encoding.

8. **Query self-attraction.** In Probe-2 the query amplitude exceeded all history codes, so the softmax concentrated on the current tap (100% of trials); future routing probes must avoid making the query a candidate that outranks history.
9. **Effective rank is not intrinsic dimension.** d_{PR} , log-trace and spectra are estimator-level descriptions of local fluctuation covariance; they are not intrinsic manifold dimensions, and no capacity conclusion follows from them.
10. **No biological equivalence.** TAN is a computational analogue; no claim of biological equivalence is made.
11. **No large-scale claim.** Results do not transfer, by construction, to transformer or LLM-scale systems.
12. **No cognition/intelligence claim.** Nothing here bears on cognition or intelligence beyond the specific mechanistic statements made.
13. **No efficiency claim.** No metabolic, energy or information-per-spike claim is made.
14. **No multi-neuron evidence.** Organizational hypotheses discussed in Section 6.6 are untested future hypotheses.
15. **Probe-1 figure/table provenance.** The repository’s single-seed diagnostic figure and CSVs (seed 20260904) predate the frozen three-seed data; the three-seed results reported here come from the frozen log and amended-audit documents (`audit_v3_amended`), as recorded in the archive index.

8 Conclusions

We set out to determine what computational structure and limitations emerge when attention-like temporal weighting is compressed into a minimal spiking dynamical unit, and to test, layer by layer, whether temporal memory, attention-like weighting, routing and semantic binding are distinct capabilities. The frozen experiments support the following conclusions, each with the evidence level stated:

- **History dependence is established (Phase 1).** The scalar membrane potential is not a sufficient state for TAN or for any window-carrying variant; the projection onto h is not a lumpable reduction. History dependence is a property of retained context, not of attention alone, and it is conditionally gated by prediction error.
- **Attention reorganizes local response geometry (Phase 2, supported with caveats).** In the matched full representation, Full TAN shows higher effective rank and covariance scale in event-locked responses than the uniform-attention control, robustly across amplitude controls; this is a statement about local fluctuation geometry, not intrinsic dimensionality, and the shared observable frame shows no TAN-specific advantage over LIF.
- **Retrieval evidence is limited (Probe 1).** Full TAN shows evidence of improved target-relevant retrieval under content-rich temporal context relative to the no-attention control, partially replicated across seeds, while a fixed delay line remains the strongest fixed-position memory baseline.

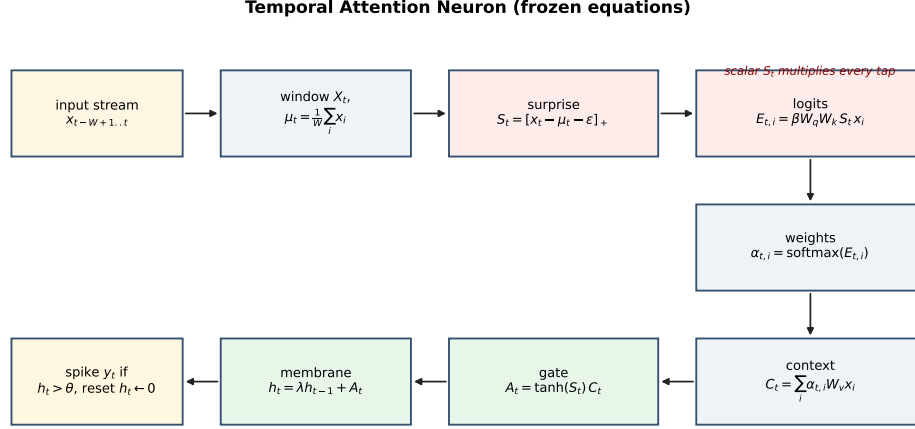


Figure 1: The frozen TAN architecture and information flow. The window $\mathcal{X}_t$ of the last $W = 5$ scalar inputs defines the local mean μ_t and the rectified surprise S_t ; logits $E_{t,i} = \beta W_q W_k S_t x_i$ share the scalar multiplier S_t ; softmax weights α form the context C_t ; the gate $A_t = \tanh(S_t) C_t$ drives the membrane, with spike/reset readout. Schematic; no data.

- **Query-conditioned routing is not implemented (Probe 2).** The scalar kernel fails the routing identifiability audit; changing the query while holding history fixed never changes which history tap is attended. This is a mechanistic identifiability result, kept in full as TERMINATED / AUDIT_FAILED in the archive.

The experiments therefore separate temporal memory, attention-like saliency weighting, routing, and semantic binding as distinct computational properties:

$$\boxed{\text{Memory} \neq \text{Weighting} \neq \text{Routing} \neq \text{Binding}},$$

and the mechanistic boundary of the current kernel can be stated precisely: *a minimal scalar temporal-attention mechanism can generate history-dependent dynamics and dynamically reorganize event-locked response structure, while remaining fundamentally limited in its ability to perform query-conditioned content selection.* Whether organizational extensions – dendritic compartments, opponent circuits, lateral or phase-based competition – lift this boundary is the legitimate next question (TAN-II); it is not answered, and not claimed, by the present paper.

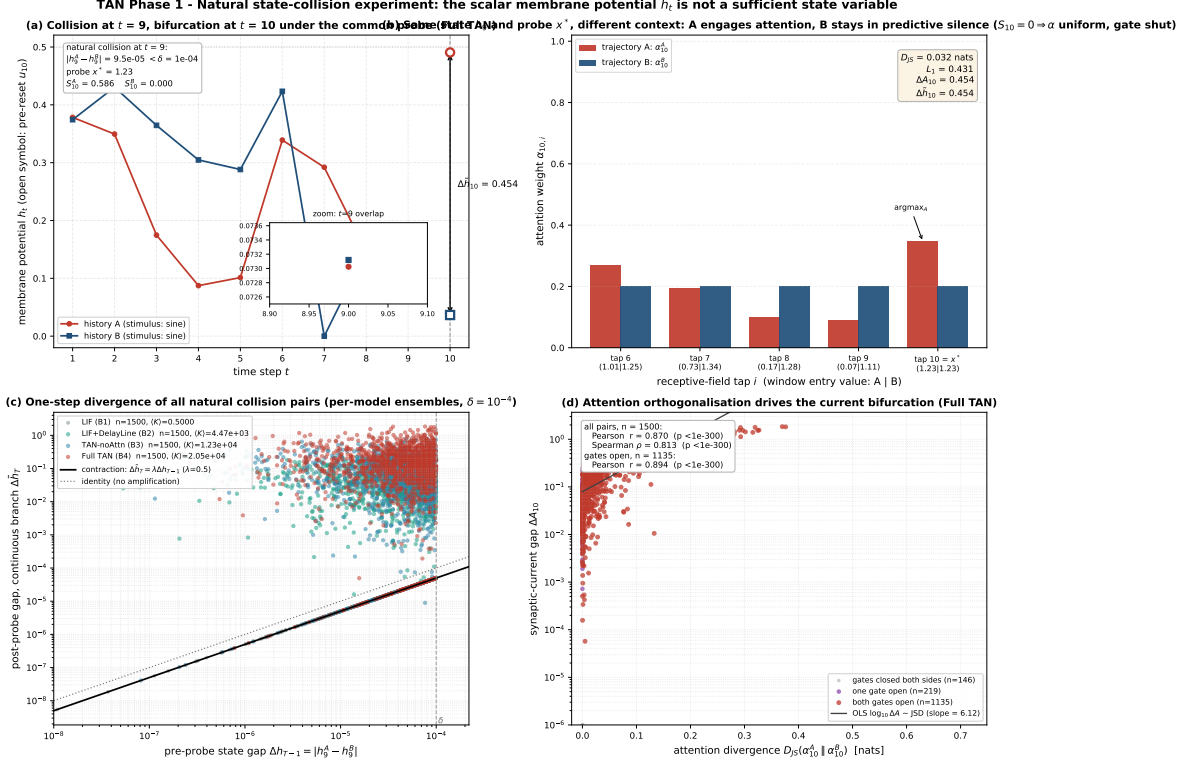


Figure 2: Phase 1, natural state collisions (frozen experiment figure `fig22_natural_collision.pdf`, four panels). (a) A representative collision pair: membrane potentials meet within $\delta = 10^{-4}$ at $t = 9$ and bifurcate at the identical probe $t = 10$ (continuous branch); (b) history-conditional attention and gate at the probe; (c) one-step divergence K of all collision pairs on log-log axes for B1–B4 (B1 lies exactly on the contraction line $\Delta h_T = \lambda \Delta h_{T-1}$); (d) attention divergence (Jensen–Shannon) vs. synaptic-current gap. Full statistics in Table 2 and the archive report.

Data and Code Availability

The frozen research archive accompanying this paper contains all experiment scripts, seeds, logs, tables, figures and audit documents: Phase 1 (`code/experiments/natural_collision.py`), Phase 2 (`code/experiments/effective_dimension.py`), Probe 1 (`code/experiments/temporal_distractor.py`), Probe 2 (`code/experiments/probe2/identifiability/`), model reference (`code/model/tan.py`), archive index (`code/RESEARCH_ARCHIVE_INDEX.md`) and reproducibility ledger (`docs/REPRODUCIBILITY_LEDGER.md`). Figures 2–6 reproduce frozen experiment figures or are drawn from frozen result documents; no new experiment was performed for this paper.

Reproducibility Statement

All experiments use deterministic seeds and frozen parameters; audit statuses (including Probe 2 = TERMINATED / AUDIT_FAILED) are archived; no post-hoc tuning was performed; per-seed tables and the archive integrity statement are provided (`audit_v3_amended/`, `docs/`). The PDF is built with the command in `paper/Makefile` (equivalently: `pdflatex main.tex`; `bibtex main`; `pdflatex main.tex`; `pdflatex main.tex`).

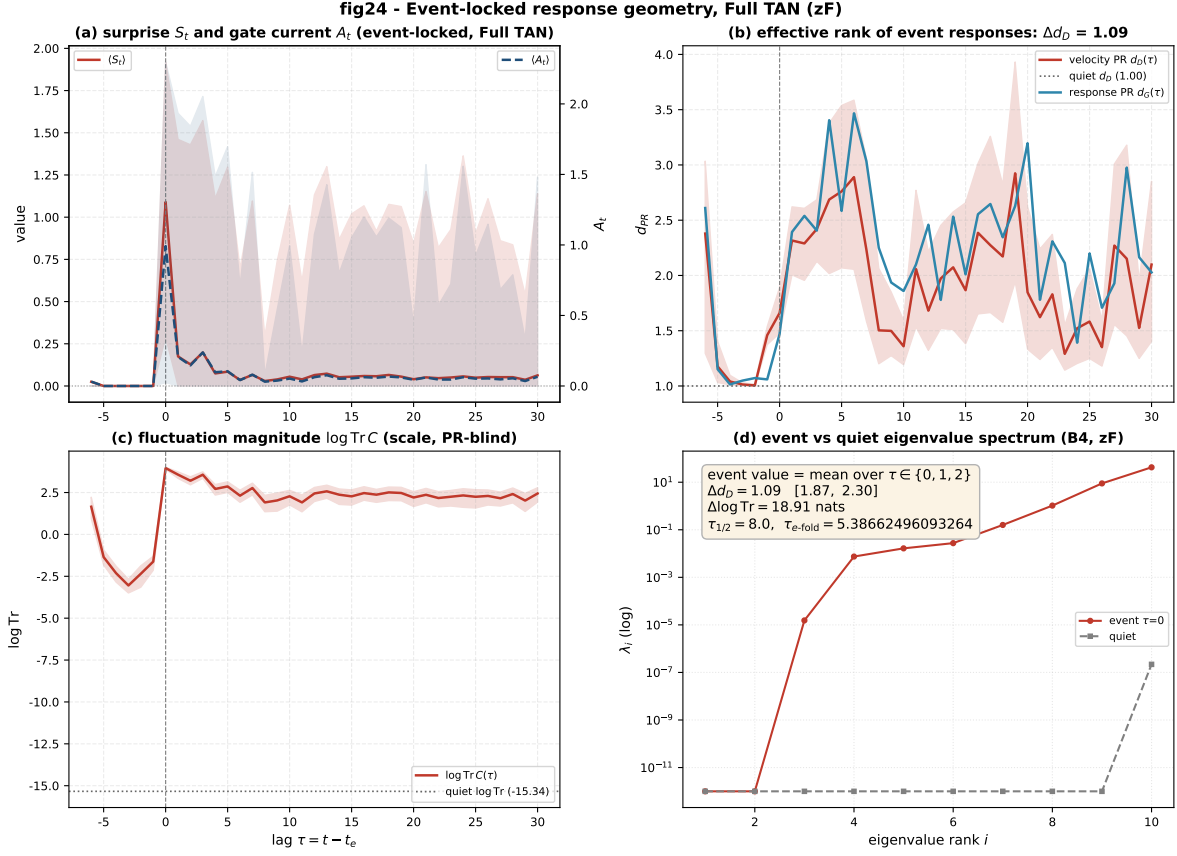


Figure 3: Phase 2, event-locked geometry of Full TAN in the full frame z_F (frozen experiment figure fig24_eventlocked.pdf, four panels): (a) event-locked surprise S_t and gated current A_t ; (b) response effective rank $d_G(\tau)$ and velocity rank $d_D(\tau)$ with quiet reference; (c) fluctuation scale $\log \text{Tr} C(\tau)$; (d) event ($\tau = 0$) vs. quiet eigenvalue spectrum. Effective rank is a covariance statistic, not an intrinsic dimension.

Conflict of Interest

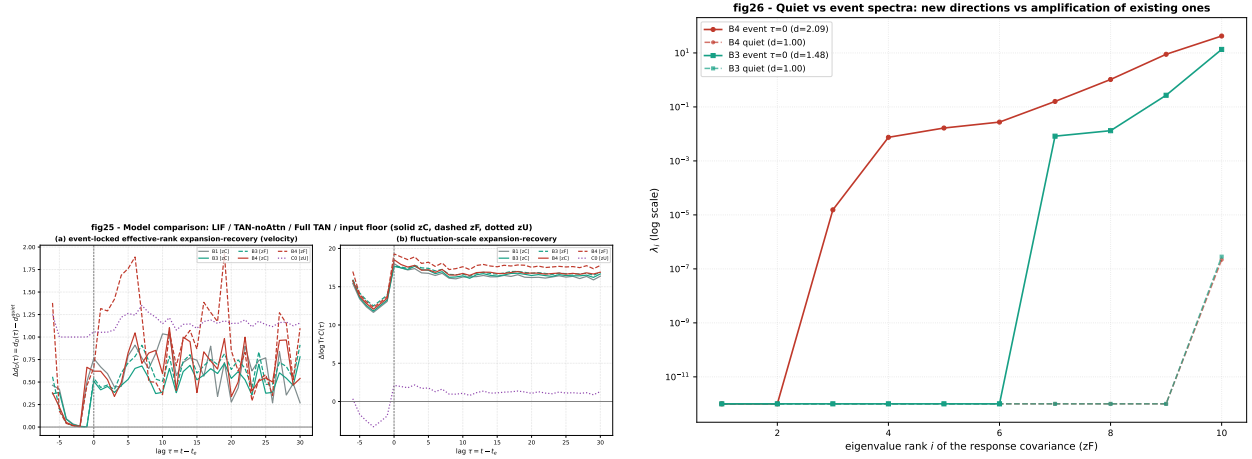
The author declares no competing interests.

Acknowledgements

The author thanks the reviewers of the archived research programme for insisting on identifiability audits and on the separation of claims documented in this paper.

References

- Larry F. Abbott. Lapicque's introduction of the integrate-and-fire model neuron (1907). *Brain Research Bulletin*, 50(5–6):303–304, 1999.
- Gary Aston-Jones and Jonathan D. Cohen. An integrative theory of locus coeruleus-norepinephrine



(a) Expansion-recovery profiles $\Delta d_D(\tau)$ and $\Delta \log \text{Tr}(\tau)$ for B1–B4 and the input control (frozen `fig25_recovery.pdf`; frames: solid z_C , dashed z_F , dotted z_U). (b) Quiet vs. event eigenvalue spectra for B4 and B3 in z_F (frozen `fig26_spectra.pdf`). Event-locked expansion adds and amplifies directions within the response covariance; see Table 3.

Figure 4: Phase 2, model comparison in effective rank and scale. B4 exceeds B3 in the matched full frame z_F with disjoint event confidence intervals and amplitude-control-robust contrasts; in the shared frame z_C B4 does not exceed B1.

function: Adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28:403–450, 2005.

Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly learning to align and translate. In *3rd International Conference on Learning Representations (ICLR)*, 2015.

Christoph Bandt and Bernd Pompe. Permutation entropy: A natural complexity measure for time series. *Physical Review Letters*, 88(17):174102, 2002.

Andre M. Bastos, W. Martin Usrey, Rick A. Adams, George R. Mangun, Pascal Fries, and Karl J. Friston. Canonical microcircuits for predictive coding. *Neuron*, 76(4):695–711, 2012.

Tom B. Brown, Benjamin Mann, Nick Ryder, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems 33*, pages 1877–1901, 2020.

Anthony N. Burkitt. A review of the integrate-and-fire neuron model: I. homogeneous synaptic input. *Biological Cybernetics*, 95(1):1–19, 2006.

SueYeon Chung and Larry F. Abbott. Neural population geometry: An approach for understanding biological and artificial neural networks. *Current Opinion in Neurobiology*, 70:137–144, 2021.

Albert Compte, Nicolas Brunel, Patricia S. Goldman-Rakic, and Xiao-Jing Wang. Synaptic mechanisms and network dynamics underlying spatial working memory in a cortical network model. *Cerebral Cortex*, 10(9):910–923, 2000.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019*, pages 4171–4186, 2019.

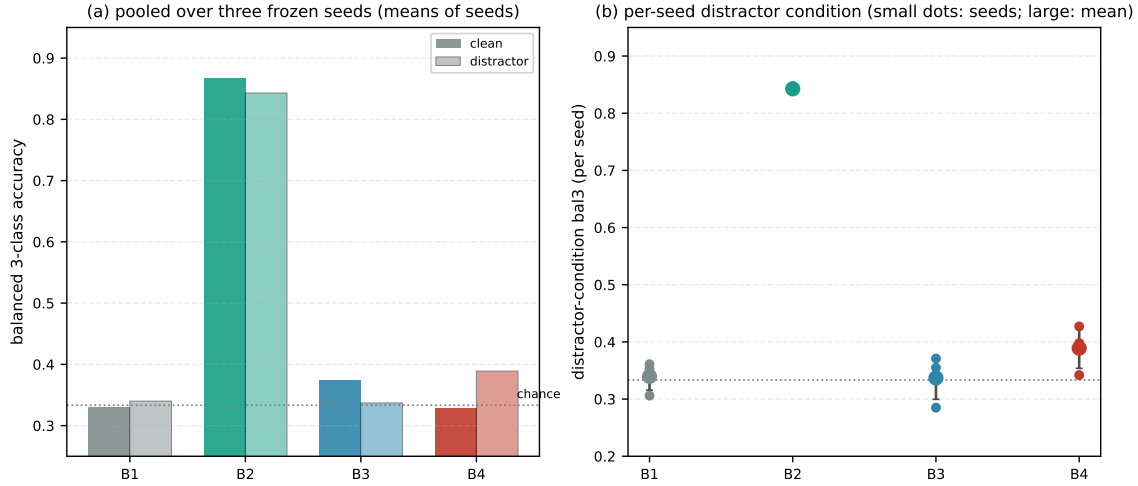


Figure 5: Probe 1 (frozen three-seed data; values from the amended audit documents `audit_v3_amended/POOLED_PROBE1_V3.md` and `audit_v3_amended/PER_SEED_AUDIT.md`): (a) pooled balanced three-class accuracy, clean vs. distractor conditions, means over three seeds; (b) per-seed distractor-condition accuracy (dots: seeds; large dot: mean). B2 is the strongest fixed-position memory baseline; B4 exceeds B3 in the distractor condition (pooled +0.0570, 95% CI [+0.0387, +0.0758]) with seed-attenuated absolute elevation; B3 has lo-blind cells (one invalid for interpretation).

Daniel Durstewitz, Jeremy K. Seamans, and Terrence J. Sejnowski. Neurocomputational models of working memory. *Nature Neuroscience*, 3:1184–1191, 2000.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. Transformer Circuits Thread (Anthropic), <https://transformer-circuits.pub/2021/framework/index.html>, 2021.

Jeffrey L. Elman. Finding structure in time. *Cognitive Science*, 14(2):179–211, 1990.

Elena Facco, Maria d’Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7:12140, 2017.

William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

Pascal Fries. A mechanism for cognitive dynamics: Neuronal communication through neuronal coherence. *Trends in Cognitive Sciences*, 9(10):474–480, 2005.

Karl Friston. A theory of cortical responses. *Philosophical Transactions of the Royal Society B*, 360(1456):815–836, 2005.

Peiran Gao and Surya Ganguli. On simplicity and complexity in the brave new world of large-scale neuroscience. *Current Opinion in Neurobiology*, 32:80–87, 2015.

Wulfram Gerstner, Werner M. Kistler, Richard Naud, and Liam Paninski. *Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition*. Cambridge University Press, 2014.

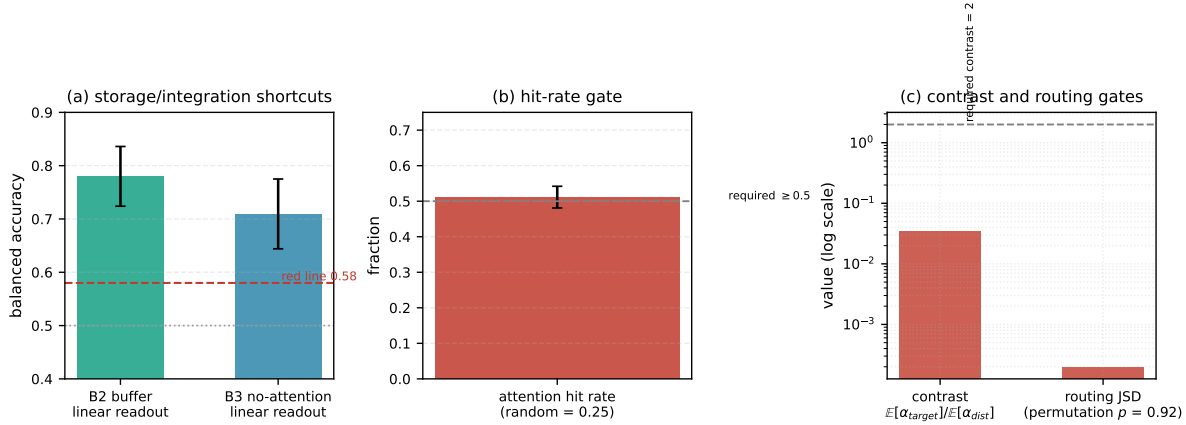


Figure 6: Probe 2 identifiability audit (values from the frozen audit documents `AUDIT_REPORT.md` / `AUDIT_OUTPUT.txt`): (a) linear-readout shortcuts of B2 (0.779) and B3 (0.709) against the 0.58 red line; (b) B4 attention hit rate 0.511 (CI [0.481, 0.542]; split $q=A$: 0.000, $q=B$: 1.000) against the 0.50 requirement; (c) target/distractor contrast $R = 0.034$ (required ≥ 2) and query-conditioned JSD 0.0002 (permutation $p = 0.92$). Audit status: TERMINATED / AUDIT FAILED (identifiability boundary).

J. Willard Gibbs. *Elementary Principles in Statistical Mechanics*. Charles Scribner’s Sons, 1902.

Patricia S. Goldman-Rakic. Cellular basis of working memory. *Neuron*, 14(3):477–485, 1995.

Peter Grassberger and Itamar Procaccia. Measuring the strangeness of strange attractors. *Physical Review Letters*, 50(5):346–349, 1983.

Alan L. Hodgkin and Andrew F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *Journal of Physiology*, 117(4):500–544, 1952.

John J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982.

Laurent Itti, Christof Koch, and Ernst Niebur. A model of saliency-based visual attention for rapid scene analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(11):1254–1259, 1998.

Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991.

Herbert Jaeger. The “echo state” approach to analysing and training recurrent neural networks. Technical Report GMD Report 148, GMD - German National Research Institute for Computer Science, 2001.

Sarthak Jain and Byron C. Wallace. Attention is not explanation. *Proceedings of the 2019 Conference of the North American Chapter of the ACL*, pages 3543–3556, 2019.

Edwin T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106(4):620–630, 1957.

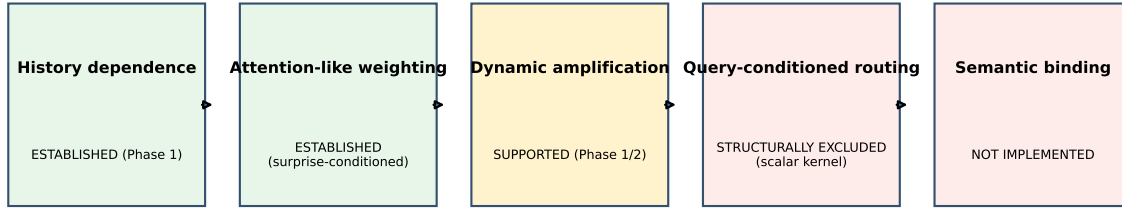


Figure 8: Mechanistic decomposition established by the frozen experiments: history dependence and surprise-conditioned weighting are established; dynamic amplification is supported; query-conditioned routing is structurally excluded by the scalar kernel; semantic binding is not implemented. Schematic; no data.

Table 1: Frozen TAN parameters (Section 3.1).

parameter	value	symbol	note
window length	5	W	receptive field
leak	0.5	λ	membrane decay
threshold	0.5	θ	spike threshold
query weight	2.0	W_q	linear map
key weight	1.0	W_k	linear map
value weight	1.0	W_v	linear map
inverse temperature	1.0	β	softmax
noise tolerance	0.0	ε	dead zone
softmax offset	10^{-9}	—	denominator
reset rule	$h_t \leftarrow 0$	—	after spike

Minh-Thang Luong, Hieu Pham, and Christopher D. Manning. Effective approaches to attention-based neural machine translation. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 1412–1421, 2015.

Wolfgang Maass, Thomas Natschäger, and Henry Markram. Real-time computing without stable states: A new framework for neural computation based on perturbations. *Neural Computation*, 14(11):2531–2560, 2002.

Luca Mazzucato, Alfredo Fontanini, and Giancarlo La Camera. Stimuli reduce the dimensionality of cortical activity. *Frontiers in Systems Neuroscience*, 10:11, 2016.

Bruno A. Olshausen, Charles H. Anderson, and David C. Van Essen. A neurobiological model of visual attention and invariant pattern recognition based on dynamic routing of information. *Journal of Neuroscience*, 13(11):4700–4719, 1993.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan,

Table 2: Phase 1: one-step continuous-branch amplification K on natural collision pairs (1,500 per model). B1 is exactly contracting ($K = \lambda = 0.5$, max deviation 1.8×10^{-8}); post-reset Markov violation rate is 0%, 0.6%, 33.1%, 32.2% for B1–B4. Source: `natural_collision_summary.csv` (frozen).

Model	n pairs	mean K	95% CI	median K	% $K > 10^3$	post-reset viol. %
B1 LIF	1500	5.000e-01	[5.00e-01, 5.00e-01]	5.00e-01	0.0	0.0
B2 LIF+delay	1500	4.470e+03	[3.26e+03, 5.92e+03]	8.96e+02	46.7	0.6
B3 TAN-noAttn	1500	1.235e+04	[7.78e+03, 1.83e+04]	1.37e+03	56.7	33.1
B4 Full TAN	1500	2.047e+04	[1.12e+04, 3.57e+04]	2.95e+03	73.7	32.2

Table 3: Phase 2: event velocity effective rank d_D (mean over $\tau \in \{0, 1, 2\}$, 374 pooled events), raw/stratified/residualised amplitude controls, event log-trace, and recovery times. Sources: `effective_dimension_summary.csv` and `effective_dimension_velocity_eventvalues.csv` (frozen). The C0 quiet reference is NaN by policy (zero variance).

Model	frame	d_D^{ev}	Δd_D	raw/strat/resid	logTr	τ_e	$\tau_{1/2}$
B1	zC	1.674	0.674	1.674/1.732/1.747	2.66	6.4	4
B3	zC	1.458	0.458	1.458/1.495/1.490	2.38	7.9	8
B3	zF	1.484	0.484	1.484/1.509/1.498	2.48	8.5	3
B4	zC	1.589	0.589	1.589/1.701/1.712	2.67	6.3	2
B4	zF	2.089	1.089	2.089/2.250/2.280	3.58	5.4	8
C0	zU	1.055	–	1.055/1.051/1.045	1.96	–	–

Ben Mann, Amanda Asbell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. arXiv:2209.11895, 2022.

Rajesh P. N. Rao and Dana H. Ballard. Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1):79–87, 1999.

John H. Reynolds and David J. Heeger. The normalization model of attention. *Neuron*, 61(2):168–185, 2009.

Sofia Serrano and Noah A. Smith. Is attention interpretable? In *Proceedings of the 57th Annual Meeting of the ACL*, pages 2931–2951, 2019.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarsz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. arXiv:1701.06538, 2017.

Wolf Singer and Charles M. Gray. Visual feature integration and the temporal correlation hypothesis. *Annual Review of Neuroscience*, 18:555–586, 1995.

Nelson Spruston. Pyramidal neurons: Dendritic structure and synaptic integration. *Nature Reviews Neuroscience*, 9(3):206–221, 2008.

Mandyam V. Srinivasan, Simon B. Laughlin, and Andreas Dubs. Predictive coding: A fresh view of inhibition in the retina. *Proceedings of the Royal Society of London B*, 216(1205):427–459, 1982.

Table 4: Probe 1 v3: per-seed balanced three-class accuracy with amended-audit statuses (24 cells; source: `audit_v3_amended/PER_SEED_AUDIT.md`, frozen). B3 20260906 clean = 0.470 is marked `INVALID_FOR_INTERPRETATION` (lo-blind); at-chance collapsed cells are `PASS_WITH_FLAG`.

seed	model	condition	bal3	coverage	status	final status
20260904	B1	clean	0.346	PASS_WITH_FLAG		PASS_WITH_FLAG
20260904	B1	dist	0.352	PASS_WITH_FLAG		PASS_WITH_FLAG
20260904	B2	clean	0.868	PASS		PASS
20260904	B2	dist	0.847	PASS		PASS
20260904	B3	clean	0.333	PASS_WITH_FLAG		PASS_WITH_FLAG
20260904	B3	dist	0.371	PASS_WITH_FLAG		PASS_WITH_FLAG
20260904	B4	clean	0.328	PASS		PASS
20260904	B4	dist	0.427	PASS		PASS
20260905	B1	clean	0.318	PASS		PASS
20260905	B1	dist	0.306	PASS		PASS
20260905	B2	clean	0.857	PASS		PASS
20260905	B2	dist	0.841	PASS		PASS
20260905	B3	clean	0.322	PASS		PASS
20260905	B3	dist	0.355	PASS		PASS
20260905	B4	clean	0.329	PASS		PASS
20260905	B4	dist	0.398	PASS		PASS
20260906	B1	clean	0.329	PASS_WITH_FLAG		PASS_WITH_FLAG
20260906	B1	dist	0.361	PASS_WITH_FLAG		PASS_WITH_FLAG
20260906	B2	clean	0.877	PASS		PASS
20260906	B2	dist	0.840	PASS		PASS
20260906	B3	clean	0.470	INVALID_FOR_INTERPRETATION		
allowbreak	PRETATION	INVALID_FOR_INTERPRETATION				
allowbreak	PRETATION					
20260906	B3	dist	0.285	PASS		PASS
20260906	B4	clean	0.329	PASS		PASS
20260906	B4	dist	0.342	PASS		PASS

Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press, 2nd edition, 2014.

Floris Takens. Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence*, volume 898 of *Lecture Notes in Mathematics*, pages 366–381. Springer, 1981.

Anne M. Treisman and Garry Gelade. A feature-integration theory of attention. *Cognitive Psychology*, 12(1):97–136, 1980.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30*, pages 5998–6008, 2017.

Christoph von der Malsburg. The correlation theory of brain function. In Eytan Domany, J. Leo van Hemmen, and Klaus Schulten, editors, *Models of Neural Networks II*, pages 95–119. Springer, 1995.

Table 5: Probe 2 identifiability audit summary (frozen). Status: TERMINATED / AUDIT_FAILED: a mechanistic identifiability boundary, not an incomplete experiment.

audit	measured	requirement / gate
B2 linear readout BA	0.779 [0.724, 0.836]	≤ 0.58 (FAIL)
B3 linear readout BA	0.709 [0.644, 0.775]	≤ 0.58 (FAIL)
attention hit rate	0.511 [0.481, 0.542]	≥ 0.50 (FAIL); $q=A$: 0.000, $q=B$: 1.000
target/distractor contrast R	0.034	≥ 2.0 (FAIL)
query-conditioned JSD	0.0002	permutation $p=0.92$ (FAIL)
counterfactual argmax flip	0.000	> 0 required (FAIL)
single-tap conditional bias	up to 0.45–0.60 at B-codes	structural note

Table 6: Claim ledger of this paper (evidence levels follow the frozen archive claim ledger, docs/CLAIM_LEDGER.md).

claim	status	evidence
h_t is a sufficient Markov state	REJECTED	Phase 1
TAN is history-dependent	ESTABLISHED	Phase 1
higher intrinsic dimension than LIF	NOT SUPPORTED	Phase 2
event-locked geometry reorganises (rank/scale)	SUPPORTED	Phase 2
fixed buffer can outperform TAN	ESTABLISHED	Probe 1
improved retrieval in content-rich context (vs no-attn)	LIMITED SUPPORT	Probe 1
distractor robustness	NOT ESTABLISHED	Probe 1
genuine Q–K routing / query-conditioned selection	REJECTED / STRUCTURALLY EXCLUDED	Probe 2 + theory
surprise-conditioned saliency weighting	ESTABLISHED	theory + all phases

Christoph von der Malsburg and Werner Schneider. A neural cocktail-party processor. *Biological Cybernetics*, 54:29–40, 1986.

Sarah Wiegrefe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 Conference on EMNLP-IJCNLP*, pages 11–20, 2019.

Angela J. Yu and Peter Dayan. Uncertainty, neuromodulation, and attention. *Neuron*, 46(4): 681–692, 2005.

A Supplementary S1: Detailed TAN equations and conventions

The frozen model used in every experiment of this paper:

$$\mathcal{X}_t = [x_{t-W+1}, \dots, x_t], \quad W = 5, \quad (16)$$

$$\mu_t = \frac{1}{W} \sum_i x_{t-W+i}, \quad x_t \in \mathbb{R}, \quad (17)$$

$$S_t = [x_t - \mu_t - \varepsilon]_+, \quad \varepsilon = 0, \quad (18)$$

$$Q_t = W_q S_t, \quad K_{t,i} = W_k x_{t-W+i}, \quad V_{t,i} = W_v x_{t-W+i}, \quad W_q = 2.0, \quad W_k = W_v = 1.0, \quad (19)$$

$$E_{t,i} = Q_t K_{t,i}, \quad (20)$$

$$\alpha_{t,i} = \frac{\exp(\beta E_{t,i})}{\sum_{j=1}^W \exp(\beta E_{t,j}) + 10^{-9}}, \quad \beta = 1.0, \quad (21)$$

$$C_t = \sum_i \alpha_{t,i} V_{t,i}, \quad (22)$$

$$A_t = \tanh(S_t) C_t, \quad (23)$$

$$h_t = \lambda h_{t-1} + A_t, \quad \lambda = 0.5, \quad (24)$$

$$y_t = \mathbb{K}[h_t > \theta], \quad h_t \leftarrow 0 \text{ if } y_t = 1, \quad \theta = 0.5. \quad (25)$$

Conventions: the window includes the current input x_t ; the receptive field is zero-padded at trial onset (stimulus onset after silence); the initial membrane potential is $h_0 = 0$; surprise is computed with μ_t over the window including x_t ; a spike occurs when the pre-reset potential exceeds θ strictly, followed by reset. The vectorised experiment implementations were parity-checked against the scalar reference implementation `code/model/tan.py` (worst deviation in h : 1.5×10^{-10} ; spike agreement exact).

B Supplementary S2: Phase 1 collision analysis details

Stimulus pool. $N = 160,000$ histories of length $T - 1 = 9$ ($4 \times 40,000$ per stimulus family: Gaussian white noise, random step, ramp, sine), seed 20260726.

Collision ensembles (per model). Pairs with both potentials in the subthreshold band $(0.05, \theta)$ at $t = 9$ and $|h_9^A - h_9^B| < \delta = 10^{-4}$: B1 92,600 pairs, B2 19,234, B3 1,633,844, B4 1,427,817; 1,500 analysed per model. The band excludes reset-induced $h = 0$ coincidences by construction.

One-step probe. An identical probe $x^* \sim \mathcal{U}(0.5, 3.0)$ was injected at $t = 10$; divergence was measured on the continuous branch ($u_{10} = \lambda h_9 + A_{10}$) because the threshold/reset is a readout discontinuity shared by all four models. Post-reset divergence and spike-outcome classes are tabulated separately.

Key statistics (frozen). Mean K : B1 0.5000000 (exact), B2 4.47×10^3 [3.26, 5.92], B3 1.235×10^4 [7.78, 1.83], B4 2.047×10^4 [1.12, 3.57] (95% bootstrap). $K > 10^3$ in 0/46.7/56.7/73.7% of pairs. Gate-closed collisions ($S_{10} = 0$, both sides): $K = 0.5$ exactly. Probe outcomes (none/one/both spiked): B1 0/0/100%; B2 0.1/0.5/99.4%; B3 38.9/7.1/54.1%; B4 33.3/8.6/58.1%; post-reset violations 0/0.6/33.1/ 32.2%. Correlations (B4): Pearson $r(\text{JSD}, \Delta A) = 0.870$ (all pairs) and 0.894 (both gates open), $p < 10^{-300}$; $r(\Delta A, \Delta \tilde{h}) = 1.000$. Cross-model log- K comparisons: B4 vs. B3 $p = 1.3 \times 10^{-32}$ (median ratio 2.15), B4 vs. B2 $p = 1.3 \times 10^{-63}$ (3.29), B2 vs. B1 $p < 10^{-300}$.

Reset re-bifurcation. $n = 843$ pairs in which all four models double-spike at $t = 10$ (identical observable state $h_{10} = 0$) continue with identical inputs: B1 gap $\equiv 0$ (machine-exact); B2 re-diverges on the continuous branch ($\langle |\Delta u| \rangle = 0.115$ at $t = 11$) and returns to zero after the window flush ($t = 14$); B3/B4 re-divergence decays but $\sim 25\%$ of pairs retain gaps $> 10^{-3}$ at $t = 18$.

Interpretation. The projection $\pi : (h, \tilde{X}) \mapsto h$ is not lumpable: on surprise-open fibres the next-state map is context-valued. History dependence is shared by all window-carrying models (B2, B3, B4); attention modulates its magnitude and the gate makes it conditional on prediction error.

C Supplementary S3: Phase 2 methodology

Observables and frames. Response trajectories are embedded as $z_U = [\mu, S, x]$ (input-only control C0), $z_C = [h, \mu, S, A]$ (models B1–B4; for B1, $A := x_t$ is a recorded null convention) and $z_F = [h, \mu, S, A, \alpha_1, \dots, \alpha_5, C]$ (B3/B4 only; for B3, $\alpha_i \equiv 1/5$ and $C = \mu$ by definition – a degenerate but not noise-filled coordinate set).

Standardisation and pooling. Coordinates are globally Z -scored per seed (post-burn-in statistics); coordinates with $\sigma < 10^{-12}$ are treated as constant and set to zero (counted, never noise-filled). No full covariance whitening is applied. Because per-seed Z -scoring assigns arbitrary per-seed levels, quiet blocks and the raw-state secondary matrices are centred per seed before cross-seed pooling; Δz and velocity quantities cancel those levels by construction, so all decision variables are unaffected (estimator note recorded in the archive log).

Event-locked geometry. Episodes are onset-aligned, $\tau \in [-6, 30]$; only events with a clean five-step pre-window are kept (374 pooled events across three seeds). Primary response: $\Delta z_{e,\tau} = z'_{e,\tau} - \bar{z}'_{e,\text{pre}}$; velocity $v_{e,\tau} = z'_{e,\tau+1} - z'_{e,\tau}$ (same-event adjacent increments). Effective rank $d = \text{Tr}^2 / \text{Tr}^2$ of the pooled covariance at fixed τ ; $\log \text{Tr}$ and eigenvalue spectra always accompany d (PR is scale-invariant and cannot detect pure amplification). NaN policy: $n < 30$ or $\text{Tr} \leq 10^{-12}$ returns NaN (never 1). Amplitude controls: raw pooled; median-split stratification with pooled-within covariance; per-coordinate linear residualisation on $\log_{10}$ amplitude.

Quiet reference and recovery. Quiet samples lie ≥ 12 steps from any pulse (mutual spacing ≥ 12). Recovery is estimated on isolated events (next onset beyond $\tau = 30$): $\tau_{1/2}$ on d_D and $\tau_{e\text{-fold}}$ from a linear fit of $\log \text{Tr}(\tau)$ (requires ≥ 0.3 nats drop), with event bootstrap 95% CIs.

Decision gates (pre-registered). Event value = mean over $\tau \in \{0, 1, 2\}$; stable expansion $\Delta d_D \geq 0.5$ with CI excluding 0. Outcome A (TAN-specific rank expansion beyond controls) requires the full-frame B4 > B3 contrast *and* a shared-frame B4 > B1 contrast; outcome B (scale/amplification channel) requires $\Delta \log \text{Tr}(\text{B4}) - \Delta \log \text{Tr}(\text{B3}) \geq 0.2$ nats; the archived gate fell to D (by exclusion) because the shared-frame B4 vs. B1 leg failed – recorded, not returned.

D Supplementary S4: Probe 1 complete per-seed results

Probe-1 v3 (frozen): target at step 5 with class c_T (lo $\sim \mathcal{U}(0.55, 1.05)$, hi $\sim \mathcal{U}(0.95, 1.45)$), three distractors (same mixture) or silence, query at step 9 (retrieval cue, not in the label), catch trials with no target (one third); label three-way (lo/hi/absent); linear 3-class softmax on the response trajectory $[u_{9..13}, y_{9..13}]$; train 4000 / test 2000 per seed; seeds 20260904/05/06. The full 24-cell per-seed audit table is in Table 4 (main text). Per-seed balanced 3-class accuracy:

seed	cond.	B1	B2	B3	B4
20260904	clean	0.346	0.868	0.333	0.328
20260904	distractor	0.352	0.847	0.371	0.427
20260905	clean	0.318	0.857	0.322	0.329
20260905	distractor	0.306	0.841	0.355	0.398
20260906	clean	0.329	0.877	0.470*	0.329
20260906	distractor	0.361	0.840	0.285	0.342

* B3 20260906 clean = 0.470 accompanies lo-class blindness in the readout; this cell is INVALID_FOR_INTERPRETATION (the nominal value is retained in the archive but is not treated as healthy 3-class decoding). At-chance cells with collapsed class coverage are flagged PASS_WITH_FLAG and excluded from claims. Pooled contrasts ($n = 6000$ test trials): B4 – B3 distractor +0.0570 (95% CI

[+0.0387, +0.0758]); per-seed differences +0.056/ + 0.043/ + 0.056. B4 absolute distractor-condition bal3: 0.427 / 0.398 / 0.342 (attenuates; seed 20260906 $\approx$ chance). B2 is the strongest model in every seed (fixed-position memory baseline). B1 $\approx$ chance. Shuffled- label controls return to chance under the amended permutation-null audit.

E Supplementary S5: Probe 1 amended-audit methodology

The three-seed Probe-1 run was blocked before pooling by two pre-registered audit-statistic gates. Per the frozen-data rule (never re-run or modify the experiment), the audit statistics were amended and re-applied to the frozen three-seed data only; no experiment parameter changed. The deterministic reproduction of the frozen pipeline was verified against the frozen per-seed numbers (maximum deviation 4.7×10^{-4} , which equals the 3-decimal rounding of the frozen table).

A1 – shuffle audit. The fixed gate on the mean over ten shuffles of the shuffled-label balanced accuracy (threshold 0.37) was abolished: L-BFGS on permuted labels has a heavy-tailed overfit distribution (n=1500: 25-shuffle mean 0.346, sd 0.079; at n=4000 the mean-of-10 sd is ≈ 0.025 ; the offending B2 seed-3 value 0.378 was $\approx +1.1\sigma$ noise, not leakage). The amended audit uses an empirical permutation null (100 label permutations per seed $\times$ model, refit each time) and classifies each cell PASS / PASS_WITH_QC_LIMITATION / INVALID_FOR_INTERPRETATION from the null mean and a z-score.

A2 – class-coverage audit. A never-predicted class no longer blocks the run. At-chance cells with a missing class are PASS_WITH_FLAG (collapse expected; excluded from claims); above-chance cells with a missing class (≥ 20 true samples) are INVALID_FOR_INTERPRETATION; a pattern persistent in $\geq 2/3$ seeds sets a model-level flag. B3’s lo-blindness occurred in 1/3 seeds by the overall criterion (seed 1 overall; seed 3 clean cell), so no persistent model-level flag is set; the affected cells carry their own statuses.

Statuses. 24-cell table in Table 4; all shuffle statuses PASS; B3 20260906 clean INVALID_FOR_INTERPRETATION; B1 at-chance cells PASS_WITH_FLAG. Pooling was performed after the amended table and preserves per-seed detail, the B3 caveat and the B4 seed-3 attenuation. Machine-readable audit decisions: (audit_v3_amended/MACHINE_READABLE_AUDIT.json).

F Supplementary S6: Probe 2 full audit results

Frozen audit (probe2.identifiability_audit.py, seed 20260904, $N = 1000$, $W = 5$, $\sigma_{\text{noise}} = 0.05$; archive: code/experiments/probe2/identifiability/). Status: TERMINATED / AUDIT_FAILED.

Generator. Position balance over the four slots (A+B counts 506/476/492/526, $\chi^2 p = 0.44$); query-label mutual information $I(q; y) \approx 3 \times 10^{-4}$ ($\chi^2 p = 0.47$); best single-slot balanced accuracy 0.555 [0.487, 0.625] (below the 0.58 gate). Structural note: at B-coded amplitude values the conditional expectation $|\mathbb{E}[V_q | x_i]|$ reaches 0.45–0.60 (e.g. $x_i = 2.25$ implies $y = +1$ on the $q = B$ half of trials), so the encoding is not strictly leakage-free by the $\mathbb{E}[V_q | x_i] \approx 0$ criterion; this is recorded as an encoding-level confound.

Audit A (B2 linear readout on $[x_{t-4}, \dots, x_t]$). Balanced accuracy 0.7789 [0.7236, 0.8358] — FAIL (red line 0.58). **Audit B (B3 uniform-integration features).** 0.7094 [0.6435, 0.7745] — FAIL. Both shortcuts follow from the asymmetric key amplitudes: the B-coded event is the largest history tap, and decoding its value is correct on the 50% of trials with $q = B$.

Audit C (B4 attention at the query step, four history taps). Hit rate 0.5110 [bootstrap 0.4810, 0.5420], split hit($q=A$) = 0.000, hit($q=B$) = 1.000 — FAIL (≥ 0.50 required at the CI bound); target/distractor contrast $R = 0.034$ — FAIL (≥ 2.0 required); query-conditioned JSD

0.0002, permutation $p = 0.92$ — FAIL ($p < 0.01$ required). Diagnostic: over the full five-tap window the argmax falls on the current (query) tap in 100% of trials (self-attraction), because query amplitudes 3.0/4.0 exceed all history codes (≤ 2.25).

Audit D (counterfactuals). Query permutation: hit 0.511 (unchanged); value permutation: 0.511; position-label permutation: 0.276 (metric sanity); *same history, both query codes*: mean per-trial JSD 0.0043 and history-argmax flip fraction **0.000**. Changing the query never changes the attention winner.

Mechanistic layers. (1) Encoding shortcut (asymmetric key amplitudes); (2) scalar-kernel limitation: with $E_{t,i} = \gamma_t x_i$ the logit ratio $E_{t,i}/E_{t,j} = x_i/x_j$ is query-independent for fixed history; (3) query self-attraction. Layer 2 survives any re-encoding while the frozen kernel is unchanged.

G Supplementary S7: Probe 2 theoretical precheck

Statement. With the frozen attention kernel $E_{t,i} = \beta W_q W_k S_t x_i$ and scalar surprise $S_t \in \mathbb{R}$, for any fixed history and any query identity q that enters only through S_t :

1. if $S_t > 0$, then $\arg \max_i E_{t,i} = \arg \max_i x_i$ (history ranking is amplitude driven);
2. the ratio $E_{t,i}/E_{t,j} = x_i/x_j$ is independent of the query identity;
3. changing q changes only the common scale $\gamma_t \propto S_t$, i.e. attention sharpness, and cannot change which history item is selected;
4. consequently $\alpha(X, q=A) = \alpha(X, q=B)$ in relative ordering for identical history X ; counterfactual argmax flip rate is predicted to be zero.

Proof sketch. At a fixed time step, $\gamma_t = \beta W_q W_k S_t$ multiplies every tap; softmax is invariant to common positive scaling of logits only up to temperature – the weights are $\alpha_i \propto \exp(\gamma_t x_i)$. For $\gamma_t > 0$, α orders by x_i ; for $\gamma_t \leq 0$ (gate closed) all logits vanish and α is uniform. Query identity changes S_t through $x_t(q)$ and μ_t , hence γ_t ; it never enters per-tap terms. The counterfactual test of the audit (identical history, $q = A \rightarrow B$, argmax flip rate = 0.000) confirms the prediction empirically. Consequence: surprise-conditioned weighting $\neq$ query-conditioned routing, independent of the scalar encoding chosen, while the frozen kernel is unchanged.

Boundary of the claim. This is a statement about the frozen kernel $E_{t,i} \propto S_t x_i$ with scalar S_t ; kernels in which the query enters per-tap (e.g. metric or 2-D representations) are outside its scope and are not analysed here.

H Supplementary S8: Reproducibility details

Environment used to generate all archived results: Python 3.12.7, numpy 1.26.4, scipy 1.13.1, matplotlib 3.9.2. Only numpy/scipy/ matplotlib are used by the experiment code. Fixed seeds and sample sizes: Phase 1 – global seed 20260726, $N = 160,000$ histories; Phase 2 – seeds 20260904/05/06, $T = 5000$ per seed, 374 pooled events; Probe 1 v3 – seeds 20260904/05/06, train 4000/test 2000 per seed (pooled test $n = 6000$); Probe 2 – seed 20260904, $N = 1000$.

Archive layout (all paths relative to the repository root): experiment scripts under `code/experiments/`; analysis scripts under `code/analysis/`; model reference `code/model/tan.py`; figures under `results/figures/` (mirrored to `manuscript/figures/`); tables under `results/tables/` (mirrored to `data/experiment_results/`); NPZ ensembles under `data/experiment_`

results/; logs under results/logs/; Probe-1 amended audit under audit_v3.amended/; Probe-2 archive under code/experiments/probe2/identifiability/; documentation under docs/ (TAN_RESEARCH_DOSSIER.md, CLAIM_LEDGER.md, REPRODUCIBILITY_LEDGER.md, NEXT_AGENT_BRIEF.md, ARCHIVE_INTEGRITY.md, PHASE4_COMPLETION_REPORT.md); archive index code/RESEARCH_ARCHIVE_INDEX.md.

Reproduction commands (recorded runtimes are from the archive logs): Phase 1: `python code/experiments/natural_collision.py` (~21–36 s); Phase 2: `python code/experiments/effective_dimension.py` (54 s) and `python code/analysis/effective_dimension_velocity_ampcontrol.py` (22 s); Probe-1 amended audit: `python code/analysis/probe1_v3_amended_audit.py` (~130 s); Probe-2 audit: `python code/experiments/probe2/identifiability/probe2_identifiability_audit.py` (< 1 s; import path note in the archive README). This paper’s figures/tables were regenerated from frozen result files by `paper/scripts/make_figs_tables.py`; no experiment was re-run for this paper.

Hash record: only the Probe-2 audit script carries a recorded SHA-256 (AABF8EBBB81E911E69117591F84E4F836CF59CF8C1B505DE53A8B7C1A88143C7); all other file hashes are marked NOT RECORDED in docs/REPRODUCIBILITY_LEDGER.md (no hashes were fabricated).